



# Thorben Foerst

DevOps & Cloud Engineering

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## Skills

### Cloud (AWS)

EKS, VPC, IAM, EC2, RDS, Secrets Manager and many more (SAA-C03 certified)

### Containers & Orchestration

Kubernetes, k3s, Amazon EKS, Docker, Docker Compose, Helm, Kustomize, Traefik Ingress Controller, Longhorn

### IaC & CI/CD

Terraform, GitHub Actions, ArgoCD (GitOps), Ansible, Git

### Monitoring & Observability

Prometheus, Grafana, Uptime Kuma

### Network & Infrastructure

MikroTik, Proxmox, TrueNAS, VLANs and subnetting, WireGuard, NetBird VPN, DHCP/DNS/firewall configuration, IPv4

### Programming

C++, Python — ability to quickly familiarise oneself with new languages and codebases

### Storage & Backup

ZFS, 3-2-1 backup strategy, RustFS

### Home & IoT

Home Assistant (Zigbee, Thread, MQTT, Matter), dedicated IoT Wi-Fi

### Hardware

Set-up of a server rack, including custom-built compute nodes, and structured LAN cabling with managed switches

## Certifications

### AWS Certified Solutions Architect – Associate

SAA-C03 – May 2026

Credly verification

## Languages

German (native language)

English (B2)

## Summary

B.Sc. student in IT Management with practical experience in setting up and operating production-grade Kubernetes infrastructures — both professionally and in my own home lab, involving networking, virtualisation and container orchestration. AWS Certified Solutions Architect — Associate. Currently writing a bachelor's thesis comparing GitOps and traditional CI/CD using ArgoCD and GitHub Actions. Aim: to start a career as a DevOps / Cloud Engineer specialising in Kubernetes.

## Education

### B.Sc. in IT Management

Oct. 2022 – expected Feb. 2027

TH Köln – Campus Gummersbach

- Bachelor's thesis: *"GitOps vs. Traditional CI/CD in a Kubernetes Context: A Comparative Evaluation Study Using ArgoCD and GitHub Actions"*

## Experience

### Software Developer (Internship)

Oct. 2025 – March 2026

Foerst GmbH

- Six-month work placement in software engineering; working across the entire tech stack, as well as gaining insights into on-premises server administration and AI development

### Software developer

Oct 2021 – Sep 2022

Foerst GmbH

- One year of full-time development prior to the start of the course, including work in C++ and the company's in-house scripting language

## Projects

### Practical Project — Kubernetes Infrastructure with Terraform

2026

TH Cologne, module in the final semester, forming the basis for the Bachelor's thesis

- Set-up and deployment of a production-ready Kubernetes infrastructure using Terraform, validated end-to-end using a test application

### Homelab – Self-hosted Kubernetes & infrastructure platform

ongoing

Personal project

- Operation of two Proxmox nodes in a custom-designed network (MikroTik routers, managed switches, custom VLANs/subnets)
- A dedicated Kubernetes cluster (one master node and one worker node per Proxmox host) for self-hosted services such as Immich, Nextcloud and Vaultwarden
- TrueNAS and Home Assistant as VMs; monitoring the environment with Prometheus and Grafana on a Raspberry Pi